MTH622:QUIZ NO.3

Quiz Start Time: 11:06 AM, 03 March 2022

Question # 1 of 5 (Start time: 11:06:36 AM, 03 March 2022)

Total Marks: 1

Kinetic energy T of a rigid body in a planar motion, is defined by

Select the correct option

Reload Math Equations

- $T = \frac{1}{2} I_O \omega^2$
- $T = \frac{1}{2} m v_c^2 + \frac{1}{2} I_c \omega^2$



Click to Save Answer & Move to Next Question

Question # 3 of 5 (Start time: 11:08:22 AM, 03 March 2022) Total Marks: 1

The Moment of inertia of a hollow sphere having mass M and radius a is given by

Select the correct option

Reload Math Equations

- $I = \frac{1}{2}Ma^2$
- $I = Ma^2$
- $I = \frac{2}{5}Ma^2$
- $I = \frac{2}{3}Ma^2$

Click to Save Answer & Move to Next Question

Question # 2 of 5 (Start time: 11:07:32 AM, 03 March 2022) Total Marks: 1

If the center of mass is at origin then the coordinate axes (xyz axes) and the principal axes of the rigid body are.....

Select the correct option

Reload Math Equations

- coincident
- perpendicular

Click to Save Answer & Move to Next Question

Question # 4 of 5 (Start time: 11:09:44 AM, 03 March 2022) **Total Marks:** 1

Consider a right circular cone of density ρ, radius a and height h is composed of elementary circular discs of small thickness each parallel to the base of the cone. Choose the z-axis as the axis of symmetry and consider a typical disc of radius r and width δz them mass of the disc is

Select the correct option

Reload Math Equations

- $\delta m = \rho 2\pi r \delta z$
- $\delta m = \rho \pi r^2 \delta z$



Click to Save Answer & Move to Next Question

Question # 5 of 10 (Start time: 03:02:20 PM, 23 February 2021)

The Moment of inertia of a solid sphere having mass M and radius a is given by

Select the correct option

- $I = Ma^2$
- $I = \frac{1}{2}Ma^2$
- $I = \frac{2}{5}Ma^2$

7 of 212

Module 136

12:10 PM



pg150

MTH622:Quiz#2

Question # 1 of 10 (Start time: 09:03:47 PM, 23 February 2021)

The Moment of inertia of a hollow sphere having mass M and radius a is given by

Select the correct option

- $I = Ma^2$
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- $I = \frac{2}{5}Ma^2$
- $I = \frac{2}{3}Ma^2$

Module 155, pg106

MTH622:Quiz#2

Question # 9 of 10 (Start time: 03:06:21 PM, 23 February 2021)

The mass of a cylindrical shell of density ρ, height h thickness dr and radi

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Usman Peer Zada
37 minutes ago
4G 66% 12:05 pm

MTH622:Quiz#2

Question # 9 of 10 (Start time: 03:06:21 PM, 23 February 2021)

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A

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Module 136

2:Quiz#2

n # 7 of 10 (Start time: 02:53:13 PM, 23 February 2

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37 minutes ago
66% 12:06 pm

MTH622:Quiz#2

Question # 4 of 10 (Start time: 02:50:38 PM, 23 February 2021)

The Moment of inertia of a solid cylinder having mass M and radius a is given by

Select the correct option

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Module 152, pg99

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Usman Peer Zada
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66% 12:05 pm
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MTH622:Quiz#2

Question # 1 of 10 (Start time: 02:57:34 PM, 23 February 2021)

The Moment of inertia of a hollow cylindrical shell having mass M and radius a is given by

Select the correct option

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Module 142, pg81

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uiz#2

3 of 10 (Start time: 03:00:19 PM, 23 February 2021)

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orrect option

m

Question # 5 of 5 (Start time: 11:11:12 AM, 03 March 2022) Total Marks: 1

If a rigid body is rotating about a fixed point on ground then its kinetic energy T is defined by

Select the correct option

Reload Math Equations

- $T = \frac{1}{2}mv_c^2 + \frac{1}{2}I_c\omega^2$
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12:48 PM • 4G 0 B/s 84
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Module 136

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pg150

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Module 155, pg106

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Module 142, pg81

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Module 152, pg99

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Question # 4 of 5 (Start time: 11:09:44 AM, 03 March 2022) **Total Marks:** 1

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Reload Math Equations

- $\delta m = \rho 2\pi r \delta z$
- $\delta m = \rho \pi r^2 \delta z$



Click to Save Answer & Move to Next Question

Question # 2 of 5 (Start time: 11:07:32 AM, 03 March 2022) Total Marks: 1

If the center of mass is at origin then the coordinate axes (xyz axes) and the principal axes of the rigid body are.....

Select the correct option

Reload Math Equations

- coincident
- perpendicular

Click to Save Answer & Move to Next Question

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Question # 6 of 10 (Start time: 09:54:50 AM, 05 August 2021)

If a rigid body is rotating about a fixed point on ground then its kinetic energy T is defined by

Select the correct option

- $T = \frac{1}{2} I_0 \omega^2$
- $T = \frac{1}{2} m v_c^2 + \frac{1}{2} I_c \omega^2$

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Question # 3 of 5 (Start time: 11:08:22 AM, 03 March 2022) Total Marks: 1

The Moment of inertia of a hollow sphere having mass M and radius a is given by

Select the correct option

Reload Math Equations

- $I = \frac{1}{2}Ma^2$
- $I = Ma^2$
- $I = \frac{2}{5}Ma^2$
- $I = \frac{2}{3}Ma^2$ ✓

Select the correct option

- the diagonal elements are zero
- the off diagonal elements are zero

B

Question # 4 of 10 (Start time: 11:45:47 AM, 05 August 2021)

If the center of mass is at origin then the coordinate axes (xyz axes) and the principal axes of the rigid body are.....

Select the correct option

- coincident
- perpendicular

A

MTH622:Quiz 3 — Quiz Start Time: 11:43 AM, 05 A

Question # 1 of 10 (Start time: 11:43:47 AM, 05 August 2021) — To

If a particle of mass 4 moving with velocity $4\hat{i}$ collides another particle of mass 5 with velocity $3\hat{i}$. Then what will be the velocity of first particle afte collision if the velocity of second particle after collision is $2\hat{i}$.

Select the correct option — Reload Math

- $\frac{15}{4}\hat{i}$

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MTH622:Quiz 3

Question # 6 of 10 (Start time: 11:47:02 AM, 05 August 2021)

Kinetic energy T of a rigid body in a planar motion, is defined by

Select the correct option

- $T = \frac{1}{2} I_0 \omega^2$
- $T = \frac{1}{2} m v_c^2 + \frac{1}{2} I_c \omega^2$

B

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MTH622:Quiz 3 — Quiz Start Time: 11:43 AM, 05 August 2021

Question # 10 of 10 (Start time: 11:49:33 AM, 05 August 2021) — Total Marks: 1

For an elastic body, the distance between two particles.....under the action of applied force

67 of 212

65% 12:08 pm

$-2\cos\omega t$

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Question # 4 of 10 (Start time: 12:57:36 PM, 05 August 2021) Tota

Consider a right circular cone of density ρ, radius a and height h is composed of elementary circular discs of small thickness each parallel to the base the cone. Choose the z-axis as the axis of symmetry and consider a typical disc of radius r and width δz them mass of the disc is

Select the correct option

Reload Math

- $\delta m = \rho 2\pi r \delta z$
- $\delta m = \rho \pi r^2 \delta z$

B

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MTH622:Quiz 3 Quiz

Question # 3 of 10 (Start time: 12:56:21 PM, 05 August 2021)

If a body rotates about some external fixed point it is called revolution

Select the correct option

- revolution
- spin

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